Yilin Wang

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Yilin Wang | Dylan-Wyl10 | 0000-0002-6002-0535

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Research Interests

Seeking **Research Scientist / Applied Scientist / ML Engineer** roles in intelligent transportation systems and autonomous driving:

- **Robotaxi & AV fleet operations** – routing, dispatching, and repositioning at network scale, with fleets doubling as mobile sensing platforms; via MILP and learning-based combinatorial optimization.
- **LLM/VLM agents & world models for AV & physical AI** – mobility planning foundation models for sim agents, world-model construction, and safety-critical scenario generation & closed-loop validation.
- **Cooperative perception & V2X systems** – from algorithms to real-world field deployment.
- **Agentic system implementation & deployment** – multi-agent architectures, tool orchestration, and agent evaluation for production workflows, from smart-city & robotaxi operations to automated research pipelines.

Education

- **Purdue University** *Aug.2021 - May 2027 (Expected)*
  Doctor of Philosophy in Civil Engineering (Transportation and Infrastructure Systems Engineering) West Lafayette, US
  ◦ Research Area: CAVs, Cooperative Perception, Physics-informed ML, Robotaxi Operations; GPA: 3.8/4.00

- **Carnegie Mellon University** *Aug.2019 - Dec.2020*
  Master of Science in Advanced Infrastructure Systems Pittsburgh, US
  ◦ Area: Smart City, Deep Learning, Natural Language Processing; GPA: 3.87/4.00

- **Tongji University** *Aug.2015 - Jun.2019*
  Bachelor of Science in Civil Engineering (Geo-technical and Underground Engineering) Shanghai, China

Working Experience

- **Inchor(SaferDrive AI)** *Dec.2025 - Jun.2026*
  Simulation Development Engineer Intern Ann Arbor, MI
  ◦ Led technical development of city-scale simulation platforms for AV testing and Physical AI (TeraSim and AgentSim; projects below), and the joint NVIDIA Metropolis demo with the City of San Jose, **selected for NVIDIA CEO Jensen Huang's GTC 2026 keynote**.

Projects

- **AgentSim: Agentic City-Level Simulation for AV Testing** *Mar.2026 - Present*
  Initiator and Lead Architect | Internship, SaferDrive AI (Inchor) | Benchmarked on the Waymo Open Dataset [Demo]
  ◦ **Scene reasoning**: LLM/VLM agents reason about a logged drive and mutate it into safety-critical variants (hard braking, cut-ins, red-light runners, pedestrian crossings).
  ◦ **Behavior simulation**: a scene behavior-simulation base built from scratch, with embedded smart sim-agent models; each variant runs in closed loop and surrounding road users react once the mutation reaches them.
  ◦ **Reconstruction and rendering**: full scene reconstruction from trajectory-only data, with a video world model rendering each variant as front-camera video (public demo).

- **TeraSim: City-Scale Traffic Simulation for AV Testing** *Dec.2025 - Mar.2026*
  Core Developer | Internship, SaferDrive AI (Inchor) | Open source [Details]
  ◦ **Behavior model**: led the integration of MotionTransformer as the foundation model of NNDE for city-scale naturalistic driving, refactored into a stateless pipeline for large-scale inference.
  ◦ **NVIDIA collaboration**: powered the NVIDIA Metropolis demo with the City of San Jose ("City Operations Agent" and digital twin); built its simulation pipeline (city-scale vehicle injection, dynamic routing, demo orchestration).

- **Cooperative Perception CAV Fleet Operation for Data Collection** *Apr.2023 - Jun.2026*
  Individual Research Project, Purdue University | Funded by NSF and USDOT (CCAT) [Details]
  ◦ **RoboSense** *(Sept.2024 - Jun.2026)*: rolling-horizon MILP that routes robotaxis for both travel time and network monitoring coverage; under revision at *Transportation Research: Part C* [S.1], INFORMS 2026 talk [C.1], TRB 2026 poster [C.2].
  ◦ **Dynamic Routing of CAVs for Improving Network Coverage** *(Apr.2023 - Aug.2024)*: multi-objective CAV routing with network coverage as one objective, solved by greedy heuristics in SUMO; TRB 2024 poster [C.3].

- **Modeling Car-Following Behavior with Physics-Informed Neural Network** *Jan.2022 - Present*
  Behavior-level reconstruction of human driving: from physics-constrained prediction to cognitively interpretable response [Details]
  ◦ **IDM-Follower** *(Jan.2022 - Feb.2024; supported by USDOT FHWA)*: physics-informed neural network that integrates the Intelligent Driver Model, robust to real-time GPS noise; *IEEE Transactions on Intelligent Vehicles* (first author) [J.2], presented at IEEE IV 2024.

- **Neural Force** *(2024 - Present)*: interpretable learned car-following response fields, compared across four naturalistic driving datasets; manuscript in preparation (co-author).

- **Cooperative Driving Automation System for Urban Areas (MMITSS)** *Sept.2021 - Jun.2026*
  Multiple Projects, Funded by USDOT FHWA, Michigan DOT (via WSP), and Leidos Inc. [Details]
  - **Smart Intersections: Paving the Way for a National CAV Deployment** *(USDOT FHWA; Lead Developer and Researcher; Oct.2023 - Jun.2026)*: redeveloped MMITSS, an open-source V2X signal priority system (SAE J2735), and built its deployments at Mcity and on Ann Arbor's Ellsworth Road corridor, where transit signal priority has served Route 6 buses since August 2025.
  - **Development of Transit Signal Priority and EV Preemption for the Intelligent Woodward Corridor Project** *(WSP, for Michigan DOT's ATCMTD project on M-1; Mentor and Developer; Jan.2024 - Jun.2026)*: implemented MMITSS for the Detroit QLine streetcar and extended it to the NTCIP 1211 operation protocol.
  - **Integration CDASim with MMITSS and Performance Analysis** *(Leidos, for USDOT FHWA's CARMA program; Developer; Sept.2024 - Apr.2025)*: integrated MMITSS into CARMA-CARLA-SUMO co-simulation; 12 pull requests merged into three FHWA repositories.
  - **Signal V2X-Based Versatile Connected Autonomous Vehicle** *(Core Team Member; 2024 Mcity 2.0 Challenge)*: connected the signal controller to the roadside V2X unit, and developed and deployed a signal-aware speed advisory algorithm on a real vehicle.

Publications

C=Conference, J=Journal, S=Under Revision

[S.1] **Yilin Wang**, Yiheng Feng* (2026). **RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring**. Under revision at *Transportation Research: Part C*. Preprint: arXiv:2609.06813.

[J.1] Hanlin Chen, Vamsi K Bandaru, **Yilin Wang**, Mario A Romero, Andrew Tarko, Yiheng Feng* (2024). Cooperative Perception System for Aiding Connected and Automated Vehicle Navigation and Improving Safety. *Transportation Research Record*, 2678(12), 1498-1510.

[J.2] **Y. Wang** and Y. Feng* (2024). IDM-Follower: A Model-Informed Deep Learning Method for Car-Following Trajectory Prediction. *IEEE Transactions on Intelligent Vehicles*, 9(6), 5014-5020.

[J.3] Ya-Dong Xue, Wei Zhang, **Yi-Lin Wang***, et al. (2023). Serviceability evaluation of highway tunnels based on data mining and machine learning: A case study of continental United States. *Tunnelling and Underground Space Technology*, 142, 105418.

[C.1] **Yilin Wang**, Yiheng Feng* (2026). **RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring**. In *INFORMS 2026 Annual Meeting*, Accepted Talk.

[C.2] **Yilin Wang**, Yiheng Feng* (2026). A Cooperative Perception Based Dynamic Vehicle Routing Framework for Urban Traffic Monitoring. In *Transportation Research Board Annual Meeting 2026*, Poster Presentation.

[C.3] **Yilin Wang**, Yiheng Feng* (2024). Dynamic Routing of Connected and Automated Vehicles for Improving Network Coverage. In *Transportation Research Board Annual Meeting 2024*, Poster Presentation.

Technical Demonstrations

- **IEEE IV 2026** (Mcity): MMITSS transit signal priority demo for symposium attendees and USDOT officials. Jun.2026
- **ITS America 2026** (Detroit, MI): MMITSS phase-2 demo for Michigan DOT and other state DOT officials. Jun.2026
- **UMTRI 60th Anniversary Symposium** (Mcity): live V2X transit signal priority demo with a transit bus. Oct.2025

Skills

- **Programming & ML**: Python, C++, MATLAB, JavaScript; PyTorch, CUDA, Hugging Face
- **Simulation & Optimization**: SUMO, CARLA, CARMA, NVIDIA Isaac Sim, Gurobi
- **Systems & Tools**: Linux, Docker, Conda, AWS, SSH; GitHub, Jira, Slack; Microsoft Office, Auto-CAD, ArcGIS

Honors and Awards

- **Honorable Mention**, Mathematical Contest in Modeling of USA, 2018
- **Silver Award** (team leader), China College Students' "Internet+" Innovation and Entrepreneurship Competition, 2018

Additional Information

Languages: English (Proficiency), Chinese Mandarin (Native) | **Interests**: Soccer, Skateboarding, Range and Gaming

References

- **Yiheng Feng**, Associate Professor, Civil and Construction Engineering, Purdue University – *Ph.D. Advisor*
- **Henry Liu**, Bruce D. Greenshields Collegiate Professor of Engineering and Director of UMTRI, University of Michigan – *Internship Supervisor; Ph.D. Committee Member*
- **Ziran Wang**, Assistant Professor, Civil and Construction Engineering, Purdue University – *Ph.D. Committee Member*

Contact details for references available upon request.